Algorithm 1: Algebraic free decompression from a random principal submatrix

Input : Eigenvalues $\{\lambda_i\}_{i=1}^{n_0}$ of an accessible random principal submatrix;
Decompression grid $1 = \tau_0 < \tau_1 < \cdots < \tau_T = \tau_\star$;
Polynomial degrees (d_z, s) and optional moments $\{\mu_p\}_{p=0}^{r}$;
Spectral grid $\{x_j\}_{j=1}^{N_x}$ and offsets $\delta_1 > \cdots > \delta_L > 0$.

Output : Target density $\widehat{\rho}_{\tau_\star}$, optionally the path $\{\widehat{\rho}_{\tau_\ell}\}_{\ell=0}^{T}$, and requested spectral observables.

// Fit the initial algebraic curve (Algorithm D.1, Appendix D)

Construct adapted contours Γ around the bulks and evaluate $\widehat{m}_1(z) \leftarrow n_0^{-1}\sum_{i=1}^{n_0}(\lambda_i - z)^{-1}$ for $z \in \Gamma$.

Fit $\widehat{P}(z,m) = \sum_{i=0}^{d_z}\sum_{j=0}^{s}\widehat{c}_{ij}z^i m^j$ using the stabilized basis and available moment constraints.

// Lift the initial physical sheet (Algorithm E.1, Appendix E)

Select an anchor $(z_{\mathrm{a}}, m_{\mathrm{a}})$ from $m(z) \sim -z^{-1}$ or $\widehat{m}_1(z_{\mathrm{a}})$, then continue the physical root on $\widehat{P}(z,m) = 0$ with tangent predictor $dm/dz = -\partial_z\widehat{P}/\partial_m\widehat{P}$, Newton correction, and adaptive subdivision, obtaining $m_1(z)$.

// Continue the physical sheet under free decompression (Algorithm F.1)

for $\ell = 1, \ldots, T$ **do**

  For each $z = x_j + \mathrm{i}\delta_k$, continue (ζ, y) from $\tau_{\ell-1}$ to τ_ℓ on $\widehat{P}(\zeta, y) = 0$, $z = \zeta - (\tau-1)y^{-1}$, using the tangent predictor $\begin{bmatrix} \partial_\zeta\widehat{P} & \partial_y\widehat{P} \\ 1 & (\tau-1)y^{-2} \end{bmatrix}\begin{bmatrix} \dot{\zeta} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 0 \\ y^{-1} \end{bmatrix}$, damped-Newton correction, and adaptive subdivision.

  Set $m_{\tau_\ell}(z) \leftarrow \tau_\ell^{-1}y$.

// Recover ρ_τ by finite-δ extrapolation (Algorithm G.1)

foreach (τ_ℓ, x_j) **do**

  Extrapolate $\{\pi^{-1}\Im m_{\tau_\ell}(x_j + \mathrm{i}\delta_k)\}_{k=1}^{L}$ to $\delta = 0$, removing the Poisson floor, and set the result to $\widehat{\rho}_{\tau_\ell}(x_j)$.

// Optional spectral observables (Appendix G)

Evolve atoms and moments from their closed evolution laws; compute edges, gaps, and cusps from the spectral-curve branch-point systems.

return $\widehat{\rho}_{\tau_\star}$, optionally $\{\widehat{\rho}_{\tau_\ell}\}_{\ell=0}^{T}$, and requested spectral observables.